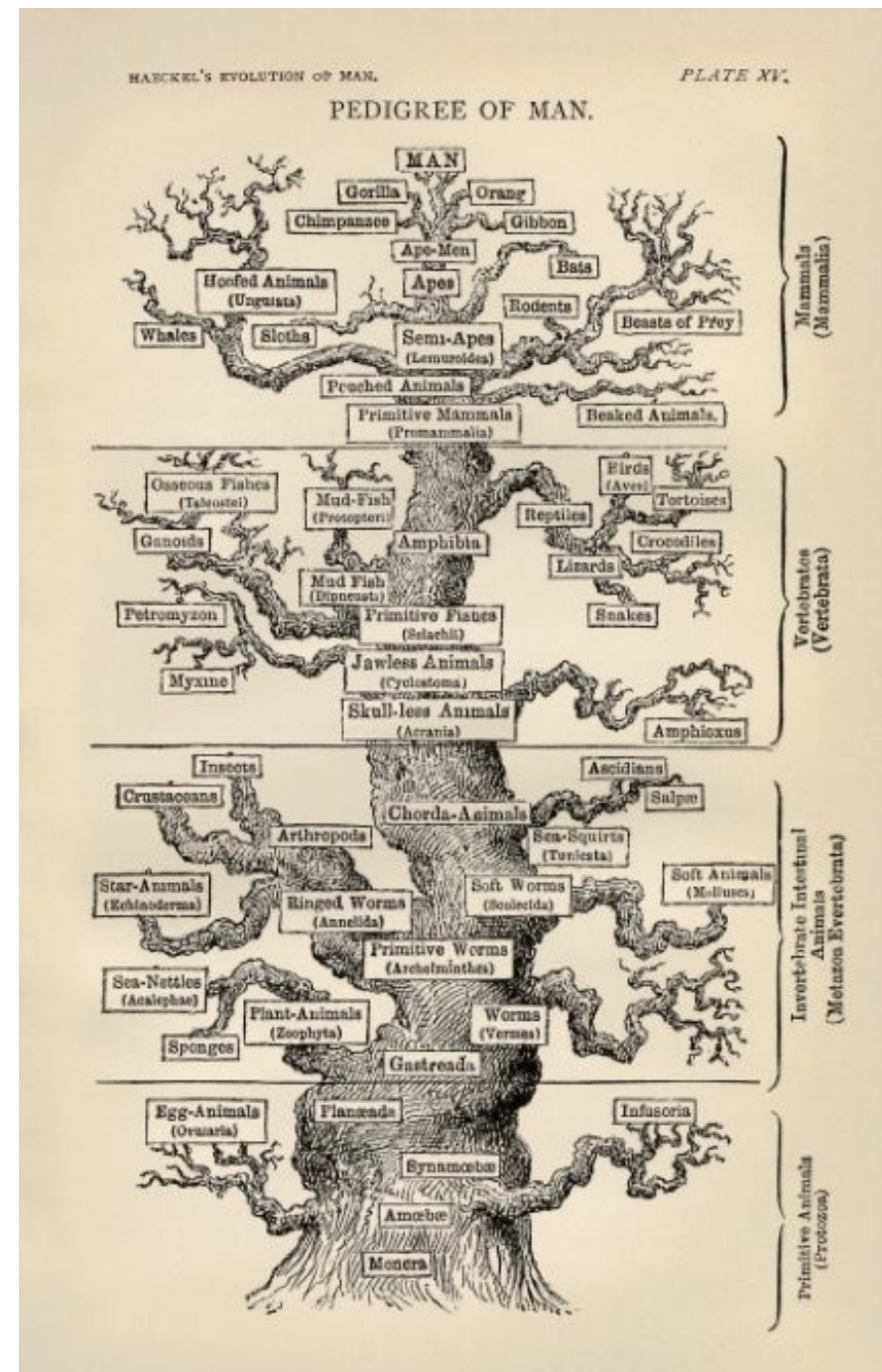
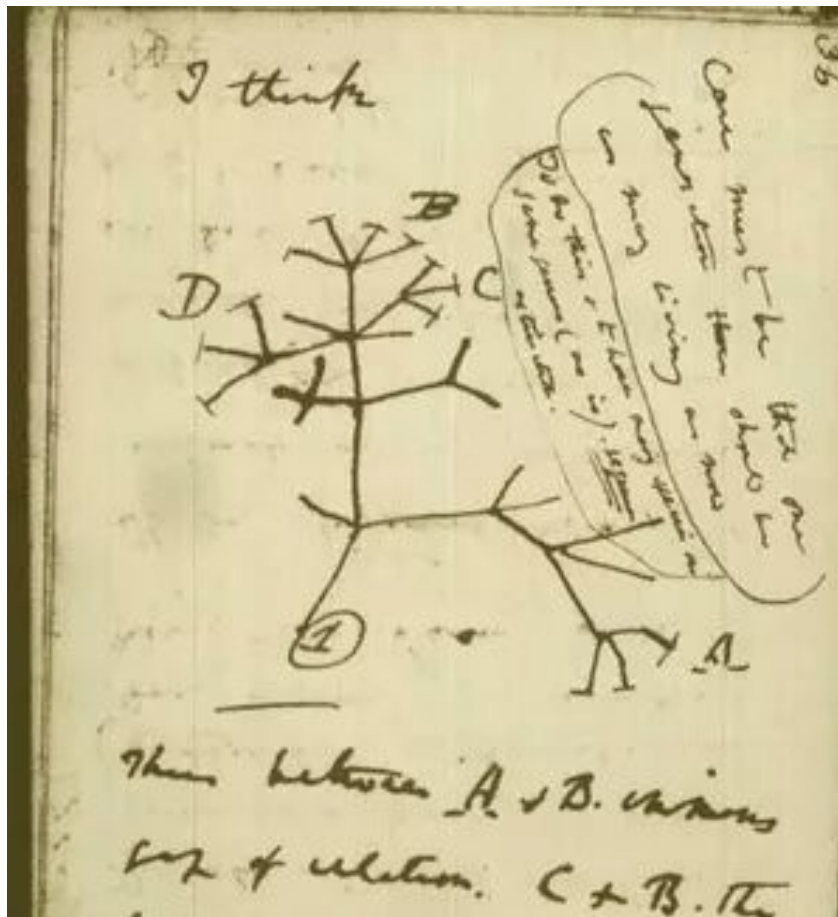
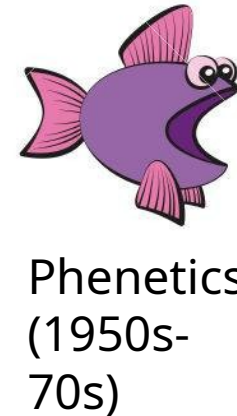
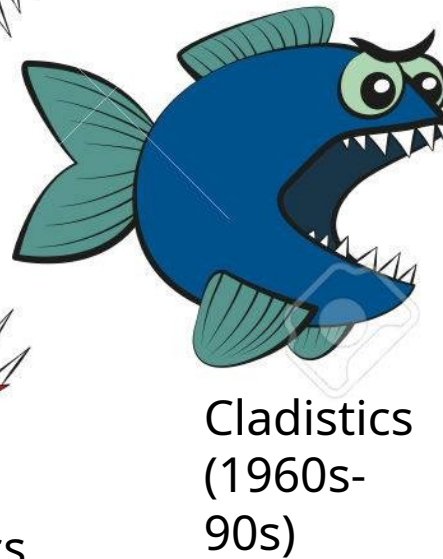
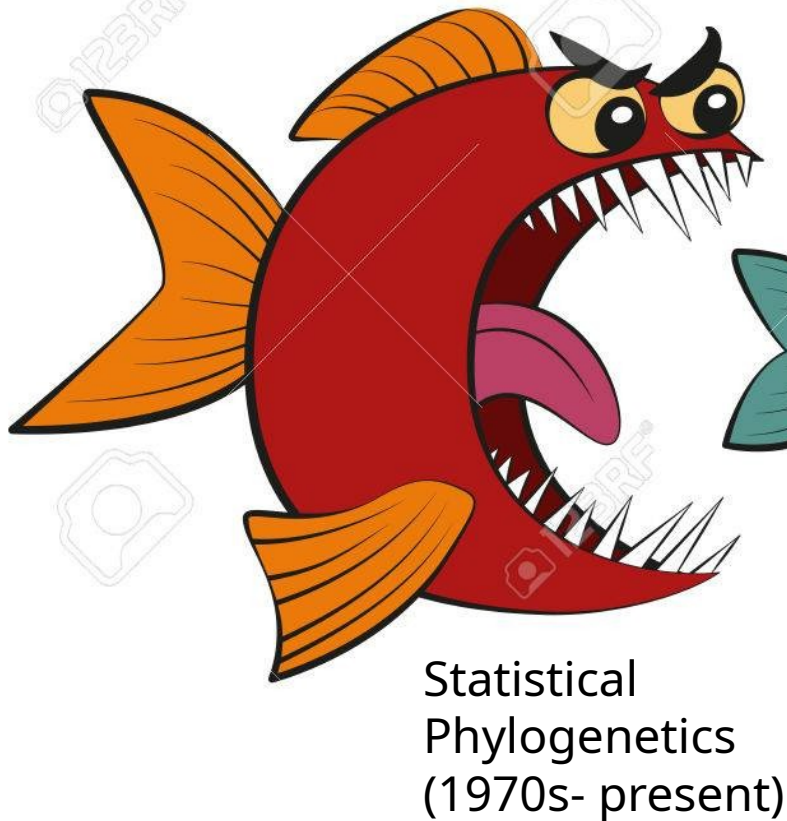


A brief history of phylogenetics



The four cannibalistic schools of phylogenetics history



History of phylogenetics: Evolutionary taxonomy

No objective methodology

Relies on expert opinion
and a lifetime of study

Strengths? Weaknesses?



History of phylogenetics: Phenetics

Numerical taxonomy and distance methods
Measure overall similarity

Sokal, Sneath, and Rohlf (1960s) argued
phenetics provided much needed
reproducibility

How do you measure similarity?

What are the drawbacks of this approach?

Distance methods

Still used for fast tree estimation (and under many conditions, are consistent estimators of the true tree)

A

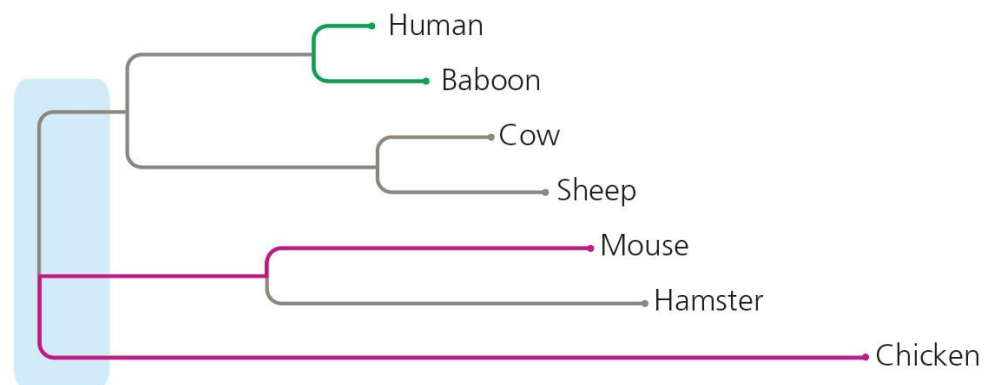
Human	M	V	H	L	T	P	E	E	K	S	A	V	T	A	L	W	G	K	V	N	D	E	V	G	G	E	A	L	G	R	L	L	V	V	Y	P	W	T	Q	R	F	F	E	S	F	G	D	L	S		
Baboon	.	.	.	.	.	.	.	.	N	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	D	.	.	.	.	.	.	
Cow	.	-	-	.	.	A	.	.	A	.	.	.	.	F	.	.	.	K	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	
Sheep	.	-	-	.	.	A	.	.	A	.	.	.	G	F	.	.	.	K	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	H	.	.	.	.	.
Mouse	.	.	.	.	.	D	A	.	A	.	.	S	G	.	.	.	.	A	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	Y	.	D	.	.	.	.	.	
Hamster	.	.	.	.	.	D	A	.	A	L	.	G	.	.	.	.	A	A	.	A	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	H	.	.	.	.	.	.	
Chicken	.	.	.	W	.	A	.	.	Q	L	I	G	.	.	.	.	A	C	A	.	.	A	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	A	.	.	.	N	.	.	.	.		

B

B

	Baboon	Cow	Sheep	Mouse	Hamster	Chicken
Human	2	6	9	8	9	13
Baboon		7	10	7	10	13
Cow			3	11	12	16
Sheep				12	9	15
Mouse					7	16
Hamster						14

C



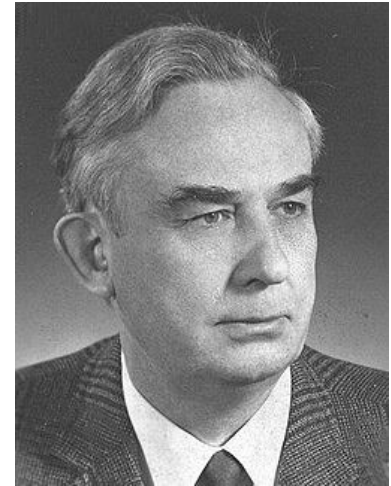
History of phylogenetics: Cladistics

Coined "phylogenetic systematics"

Grouping should not be based on *overall* trait similarity, **but only on *shared derived traits* (synapomorphies)**

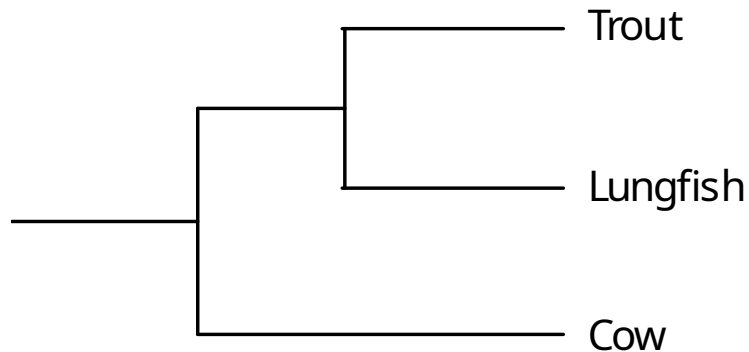
Taxonomy based on *monophyly*

Parsimony & Occam's Razor- The simplest explanation is probably the correct one

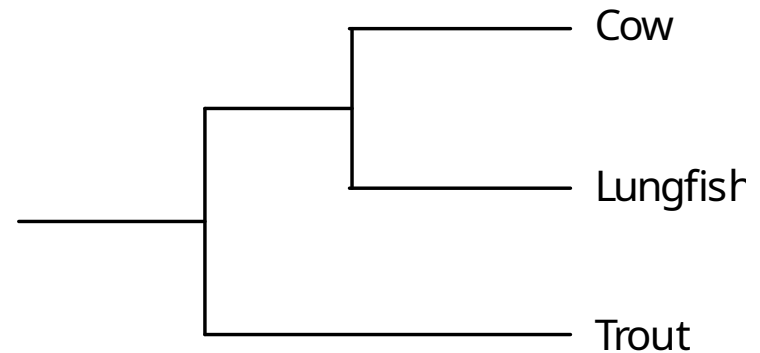


Willi Hennig (1966)

Phenetics didn't always have the goal of finding the "true" evolutionary history



"Phenogram"

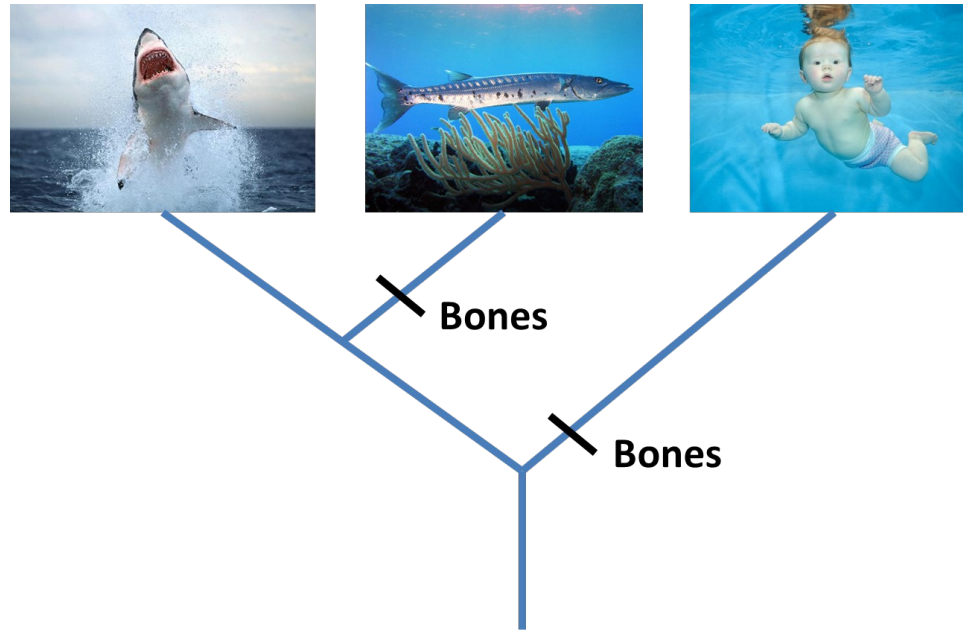


Cladogram

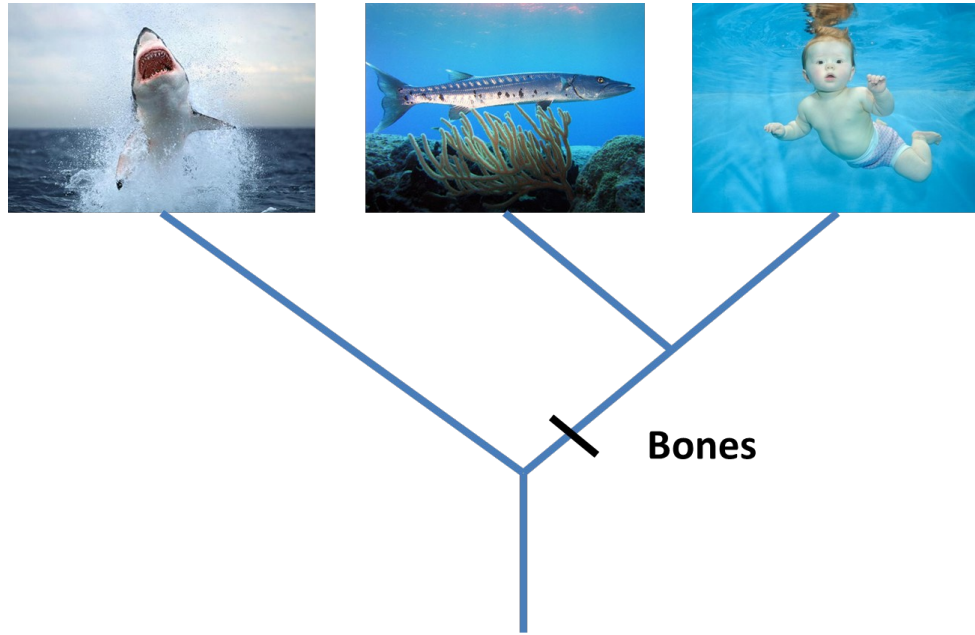
Example: What's the phylogenetic relationship?



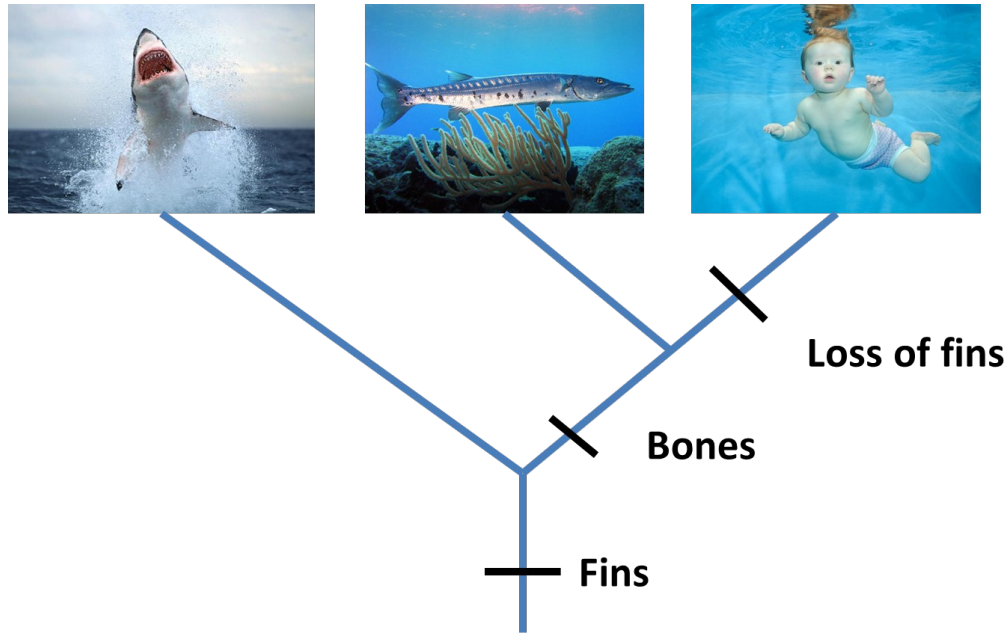
Example: Mapping bones



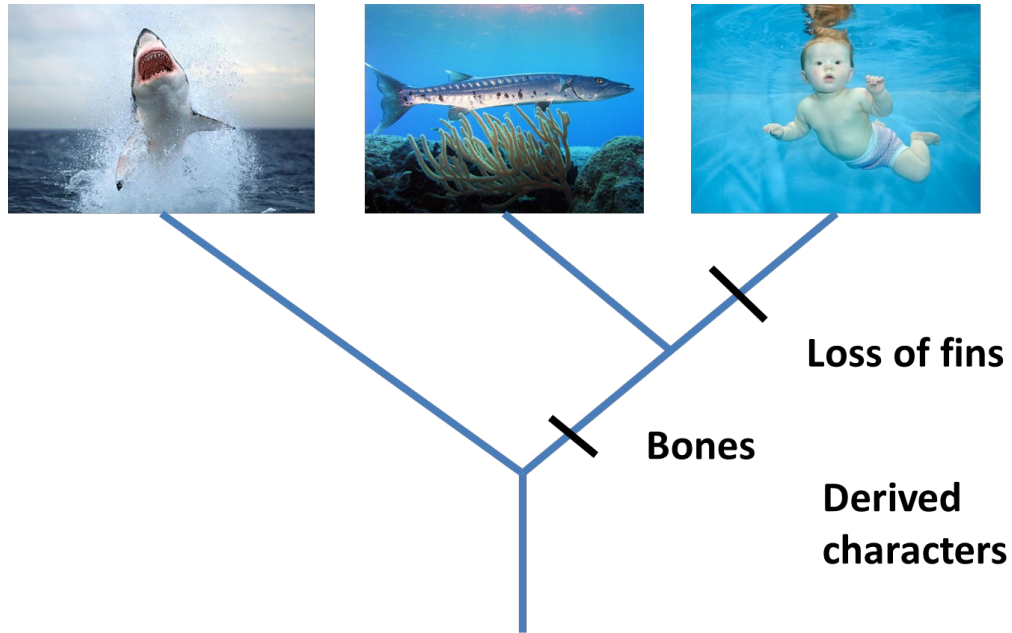
Example: More parsimonious



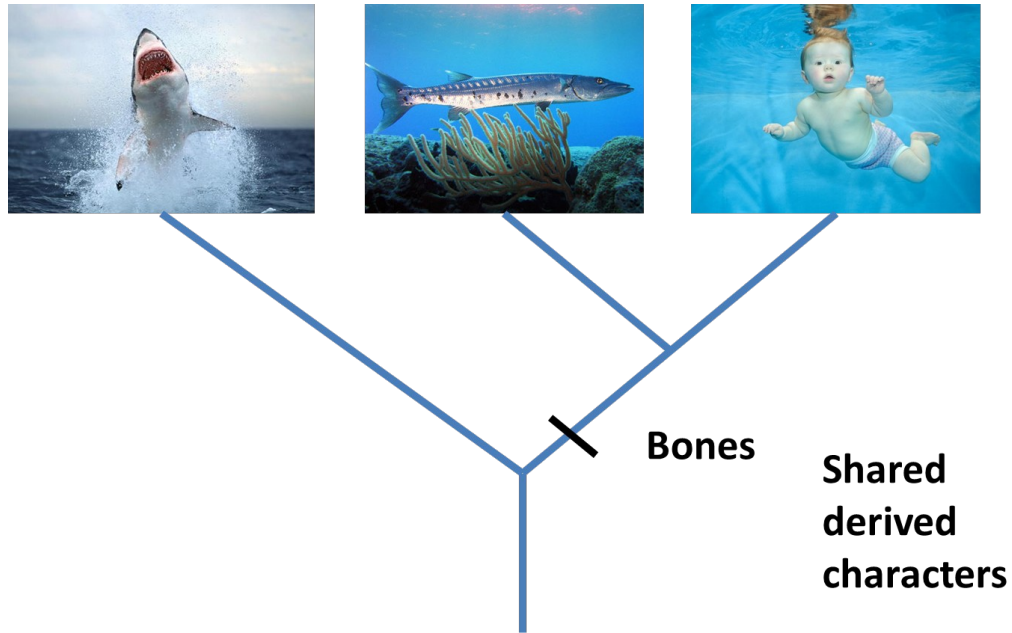
Example: Adding characters



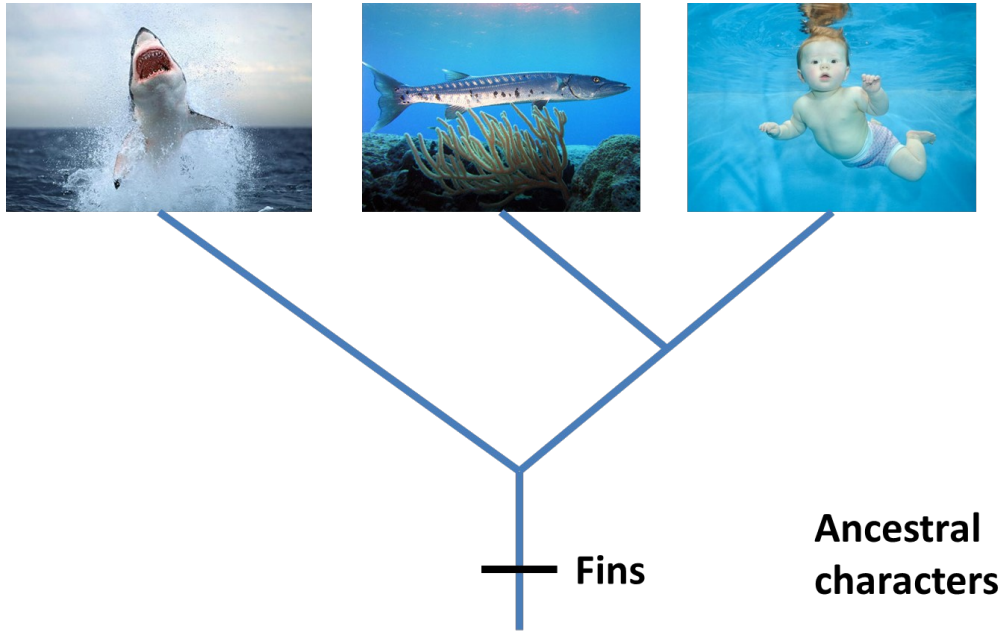
Derived Characters = Apomorphies



Shared derived characters = Synapomorphies



Shared ancestral characters = Sympleisiomorphies



Comparing similarity vs. cladistics

Overall similarity takes all characters equally

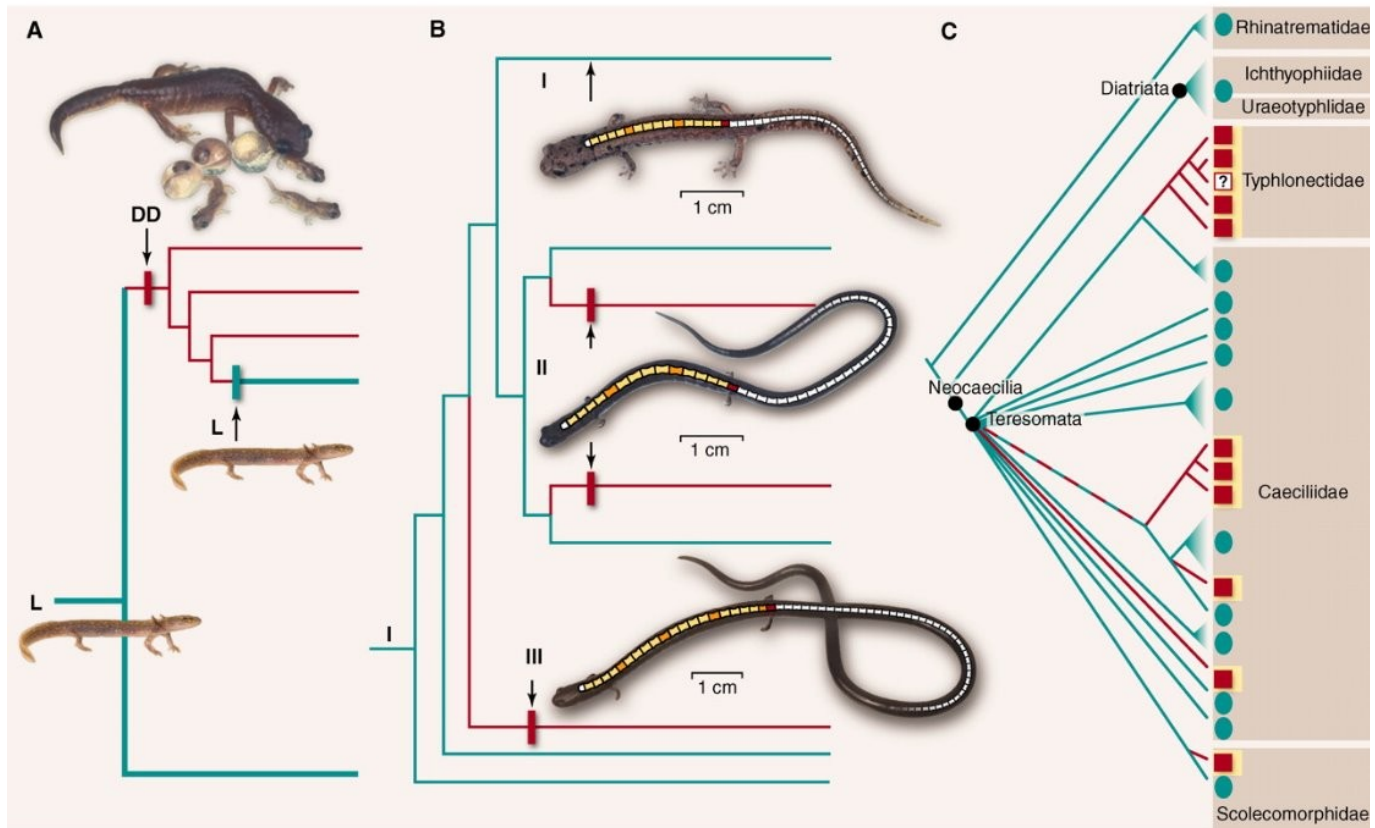
Cladistics only considers synapomorphies, and excludes autapomorphies & symplesiomorphies

Best supported hypothesis: Minimizing steps

Critical role of outgroup to distinguish derived vs. ancestral characters

Also note that characters are defined as derived/ancestral relative to the taxa being considered (the ingroup)

Homoplasy – Evolutionary reversals and/or convergence



Parsimony informative vs. uninformative

Must have at least 2 taxa with shared traits in 2 different states.

TABLE 7.7 Examples of informative and uninformative characters, with ? used to represent uncertain or missing character states.

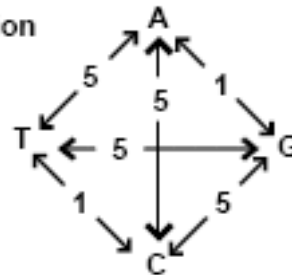
Taxon	Characters													
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	1	0	1	0	?	1	0	1	0	0	0	1	0
3	0	1	1	2	1	0	2	0	1	1	0	0	2	0
4	1	1	1	1	?	1	2	0	1	0	1	1	2	1
5	1	1	2	3	1	1	0	0	1	2	?	2	2	2
6	1	0	2	2	1	?	1	0	1	0	?	3	2	?

Generalized Parsimony

Step matrices

To		Ordered				Unordered				Dollo				Camin-Sokal			
		0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3
F	0	-	1	2	3	-	1	1	1	-	M	2M	3M	-	1	2	3
r	1	1	-	1	2	1	-	1	1	1	-	M	2M	i	-	1	2
o	2	2	1	-	1	1	1	-	1	2	1	-	M	i	i	-	1
m	3	3	2	1	-	1	1	1	-	3	2	1	-	i	i	i	-

Transition/Transversion
Parsimony



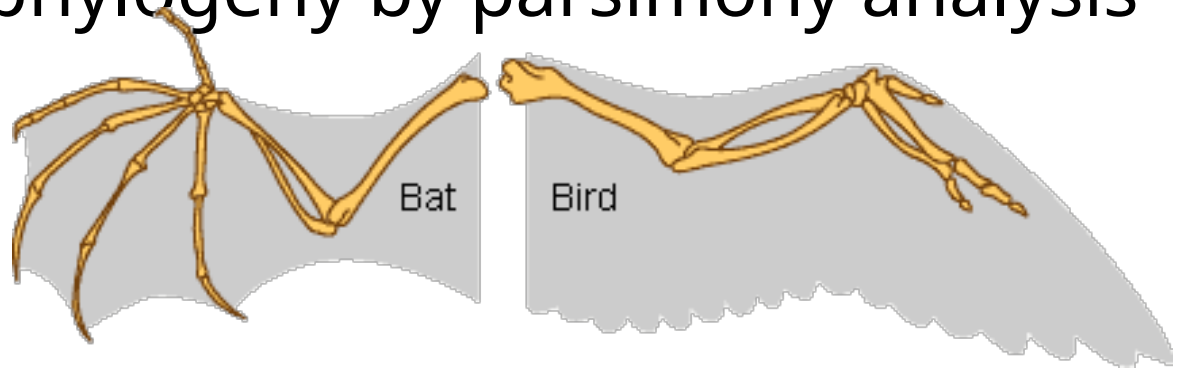
	A	C	G	T
A	-	5	1	5
C	5	-	5	1
G	1	5	-	5
T	5	1	5	-

By Peter Forey, [Link to source](#)

Homology

Primary homology = Initial hypothesis of homology due to similarity

Secondary homology = Implied homology via single origin on phylogeny by parsimony analysis



Underlines importance of character construction - often surprisingly subjective

"Reciprocal illumination" Hennig 1966

History of phylogenetic inference: Statistical phylogenetics

Luca Cavalli-Sforza, Anthony Edwards and later, Joseph Felsenstein

Probabilistic models of trait evolution

Showed statistical issues with parsimony, e.g. "the Felsenstein zone"

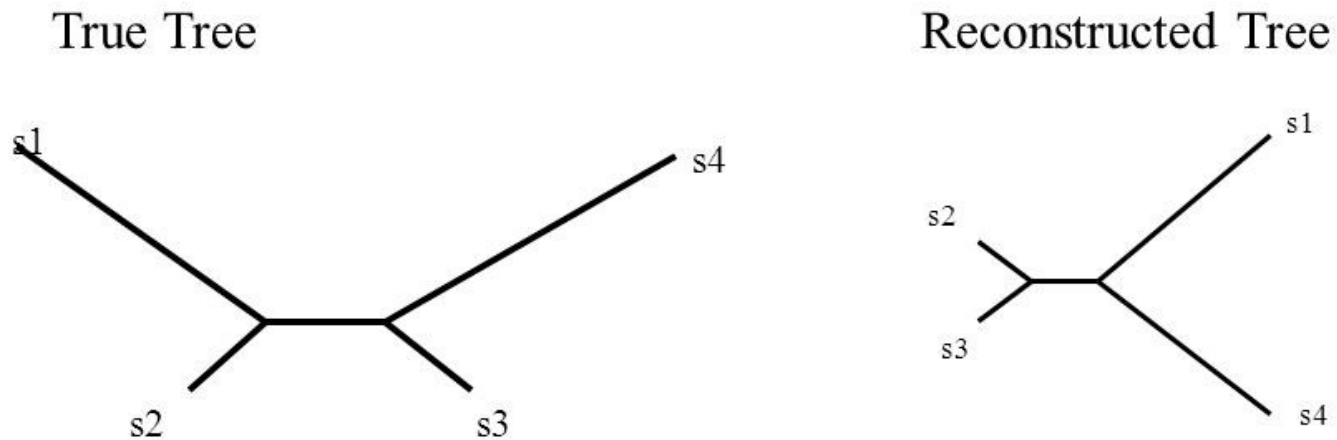
-> Highlighted the importance of simulation!

Maximum Likelihood & Bayesian inference



Joe Felsenstein

The Felsenstein Zone and Long-Branch Attraction



Lack of statistical consistency of parsimony.
Consistency: An estimator converges on the truth as more data are added

An oversimplification of the problem...

Under the cladistics philosophy (model), homoplasy is a “mistake” in character construction

Coding independently evolved traits the same, look closer and you'll see they're different!! (e.g. wings of birds & wings of bats)

But for DNA, complete convergence is real (a “T” is a “T” is a “T”, no matter how it became that)

As similarity by convergence becomes more common than similarity by shared ancestry, parsimony fails, and no “looking closer” at the character will save it.

Parsimony's main assumption of "Occam's Razor" had unclear implications

What are we assuming about the evolutionary process?

Rare changes or "no common mechanism"
(Tuffley and Steele 1997)

Inconsistent estimator in real datasets with homoplasy – defined as an estimator that gets more and more certain of the wrong answer as more data is added.

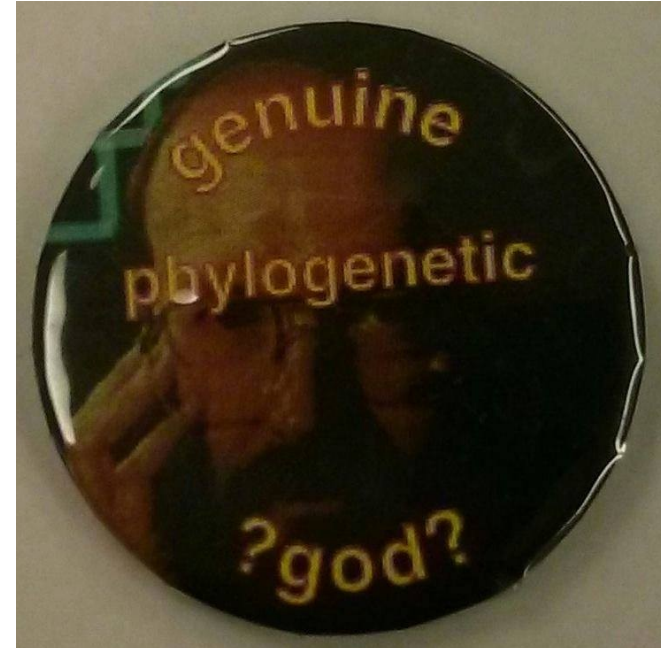
A history of conflict in science

MATT SIMON SCIENCE 02.03.16 07:00 AM

TWITTER NERD-FIGHT REVEALS A LONG, BIZARRE SCIENTIFIC FEUD



#ParsimonyGate
e



Parsimonygate

The epistemological paradigm of this journal is parsimony. There are strong philosophical arguments in support of parsimony versus other methods of phylogenetic inference (e.g. Farris, [1983](#)).

The high citation index of *Cladistics* shows that the journal is publishing some of the most ground-breaking empirical and theoretical research on the history of life, and we remain committed to the publication of outstanding systematics research. As a community of scientists, the Willi Hennig Society is always open to new methods and ideas, and to well-reasoned criticisms of old ones. However, we do not hold in special esteem any method solely because it is novel or purportedly sophisticated.

Phylogenetic data sets submitted to this journal should be analysed using parsimony. If alternative methods are also used and there is no difference among the results, the author should defer to the principles of the Society and present the tree obtained by parsimony. Unless there is a pertinent reason to include multiple trees from alternative methods, a tree based on parsimony is sufficient as an intelligible, informative and repeatable hypothesis of relationships, and articles should not be cluttered with multiple, often redundant, trees produced from other methods. If alternative methods give different results and the author prefers an unparsimonious topology, he or she is welcome to present that result, but should be prepared to defend it on philosophical grounds.

In keeping with numerous theoretical and empirical discussions of methodology published in this journal, we do not consider the hypothetical problem of statistical inconsistency to constitute a philosophical argument for the rejection of parsimony. All phylogenetic methods, including parsimony, may produce inconsistent or otherwise inaccurate results for a given data set. The absence of certain truth represents a philosophical limit of empirical science.

Cladistics will publish research based on methods that are repeatable, clearly articulated and philosophically sound. We believe these guidelines implement the vision of Willi Hennig ([1965](#), p. 97), who said, "(i)nvestigation of the phylogenetic relationship between all existing species and the expression of the results of this research in a form which cannot be misunderstood, is the task of phylogenetic systematics."

Ultimately, parsimony is less an inflexible modeling paradigm

As pointed out in the policy, other models can be inconsistent if the model does not fit or is not adequate for the data-generating process.

But statistical phylogenetics gave a way to parameterize processes, connect them to knowledge from population genetics, and test for adequacy and fit in ways that parsimony did not.

This ultimately led to statistical phylogenetics being the dominant paradigm in phylogenetics from sequence data.

Parsimony still commonly used for:

Morphological analyses (though probabilistic models exist here too)

Starting trees

Simplified reconstruction of the history of trait change (ancestral state reconstructions)

Finding the best tree: Exhaustive searches

How do you find the most
parsimonious tree?

One way:

1. Try every possible tree.
2. For each tree, count min # of
steps
3. Choose the tree with the fewest #
steps

Too many trees.

Table 3.1: The number of rooted, bifurcating, labeled trees for n species, for various values of n . The numbers for more than 20 species are approximate.

Species	Number of trees
1	1
2	1
3	3
4	15
5	105
6	945
7	10,395
8	135,135
9	2,027,025
10	34,459,425
11	654,729,075
12	13,749,310,575
13	316,234,143,225
14	7,905,853,580,625
15	213,458,046,676,875
16	6,190,283,353,629,375
17	191,898,783,962,510,625
18	6,332,659,870,762,850,625
19	221,643,095,476,699,771,875
20	8,200,794,532,637,891,559,375
30	4.9518×10^{38}
40	1.00985×10^{57}
50	2.75292×10^{76}

Heuristic searches

Don't try every tree.

"Hill-climbing algorithm"

Wander around under certain rules.
Hope that what you eventually stop at is
the best.

Not guaranteed to find the best tree.

Can be improved by lots of random
starting points.

(Not necessary for parsimony, exact
algorithmic solution possible with a
"branch and bound" search)

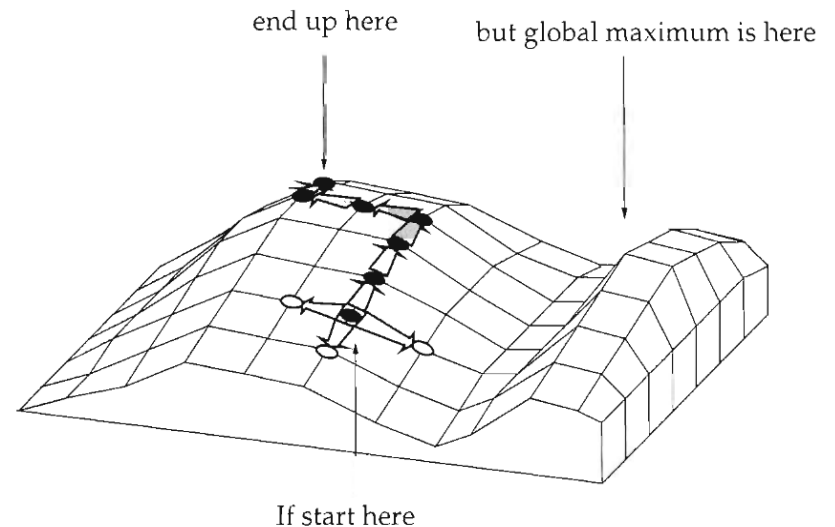


Figure 4.1: A surface rising above a two-dimensional plain (or plane). The process of climbing uphill on the surface is illustrated, as well as the failure to find a higher peak by this "greedy" method.

Tree space

Each point on the grid is a possible tree.

Trees next to each other are “1 step away” via algorithmic manipulation

Start at a random tree.

Try all trees that are nearby, choose the one that’s best

Keep doing that until you can’t make the tree any better.

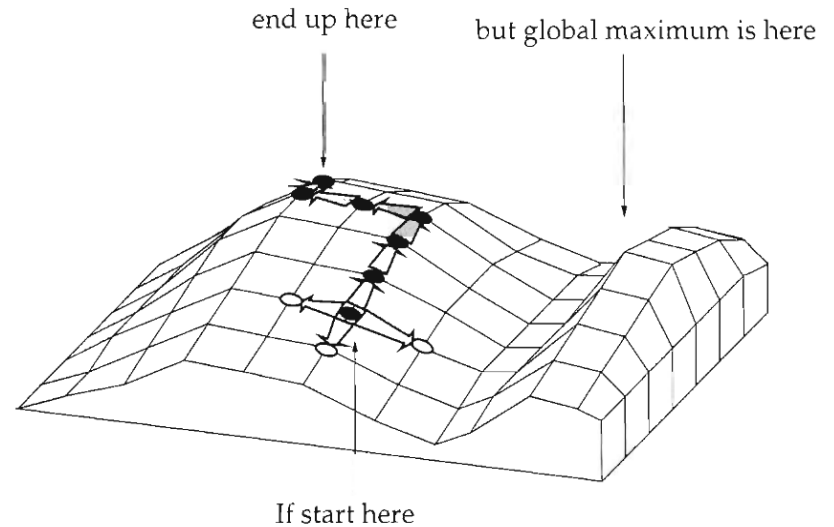


Figure 4.1: A surface rising above a two-dimensional plain (or plane). The process of climbing uphill on the surface is illustrated, as well as the failure to find a higher peak by this “greedy” method.

Visualizing tree space

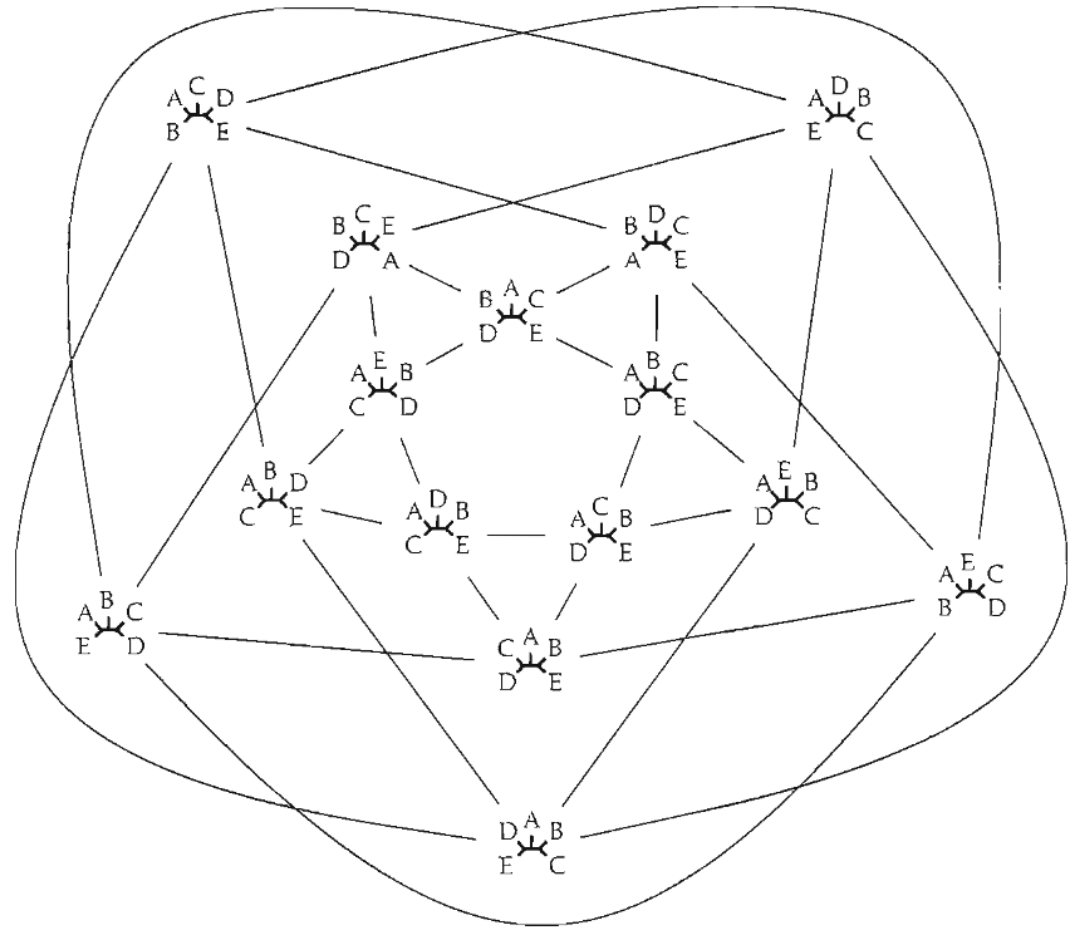


Figure 4.3: The space of all 15 possible unrooted trees with 5 tips. Neighbors are connected by lines when a nearest-neighbor interchange can convert one into the other. The labels A–E correspond to the species names Alpha through Epsilon in that data set. This symmetric arrangement of nodes was discovered by Ben Rudd Schoenberg (personal communication), and we thus denote this graph the Schoenberg graph.

Finding the shortest tree in treespace

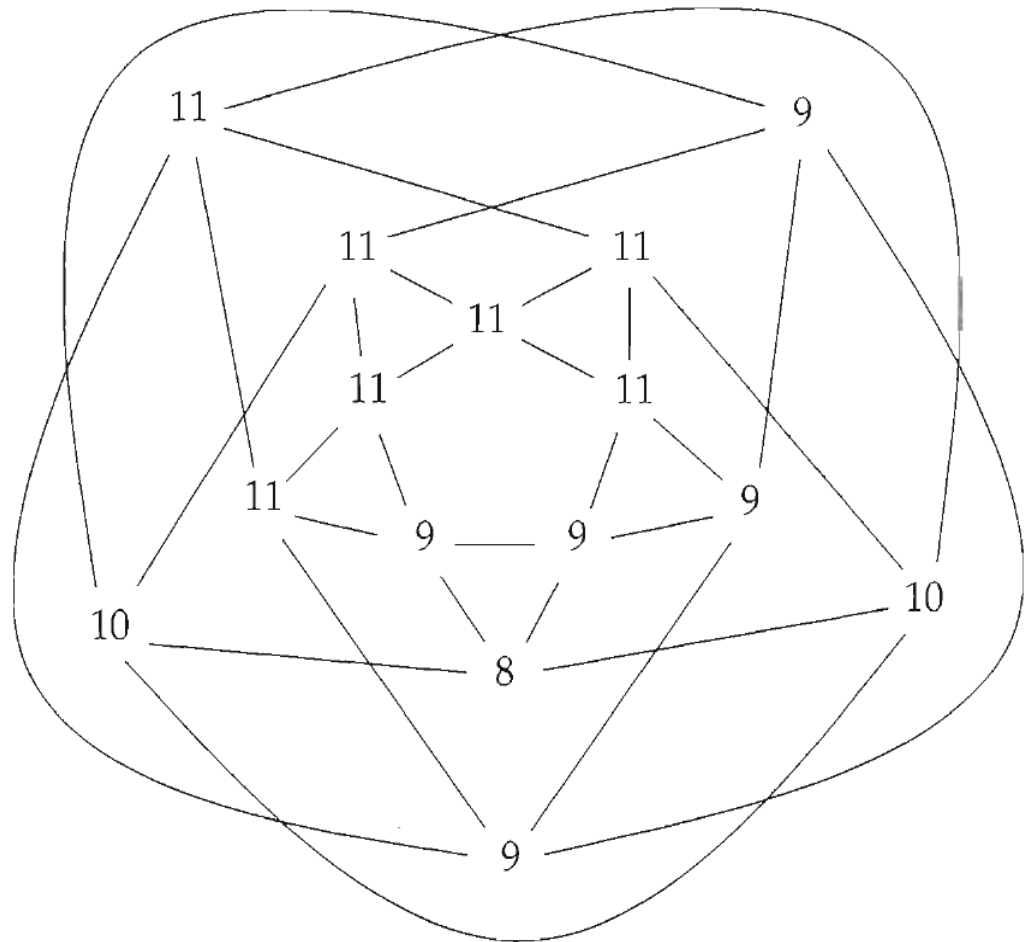
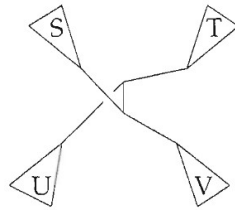
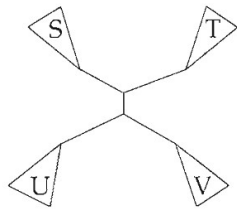
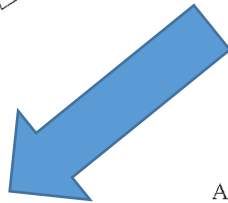


Figure 4.4: The space of all 15 possible trees, as in Figure 4.3, where the number of changes of state on the data set of Table 1.1 is shown. Nearest-neighbor interchanges search for the most parsimonious tree by moving in this graph.

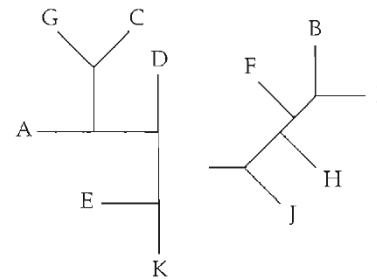
Tree space connectivity is defined by "moves"



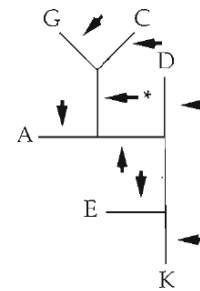
Nearest Neighbor Interchange (NNI)



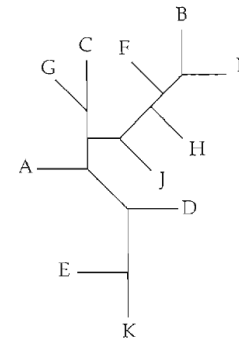
Break a branch, remove a subtree



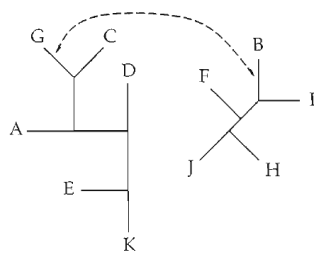
Add it in, attaching it to one (*) of the other branches



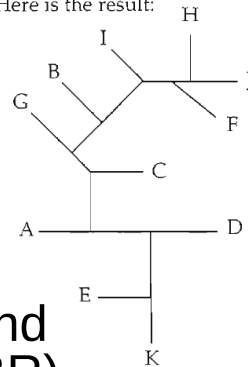
Here is the result:



Connect a branch of one to a branch of the other



Here is the result:



Tree bisection and reconnection (TBR)

Subtree pruning and regrafting (SPR)

Distance between tree spaces
changes depending on which
moves you use

NNI fastest, least computationally intensive,
but the largest distances between trees

TBR slowest, most computationally intensive,
but shortest distances between trees

SPR intermediate

Navigating tree space

Optimization/gradient descent (here ascent) is a general problem in computational analyses

It is always a balance of speed (NNI faster than SPR faster than TBR) and how efficiently you move across tree space (TBR connects more trees than SPR which is more than NNI)

Unless you sample all trees, a heuristic search may not be sure you ever find the global peak.

Trying lots of different starting points can increase your confidence.

Parsimony has an exact algorithm

Branch and bound

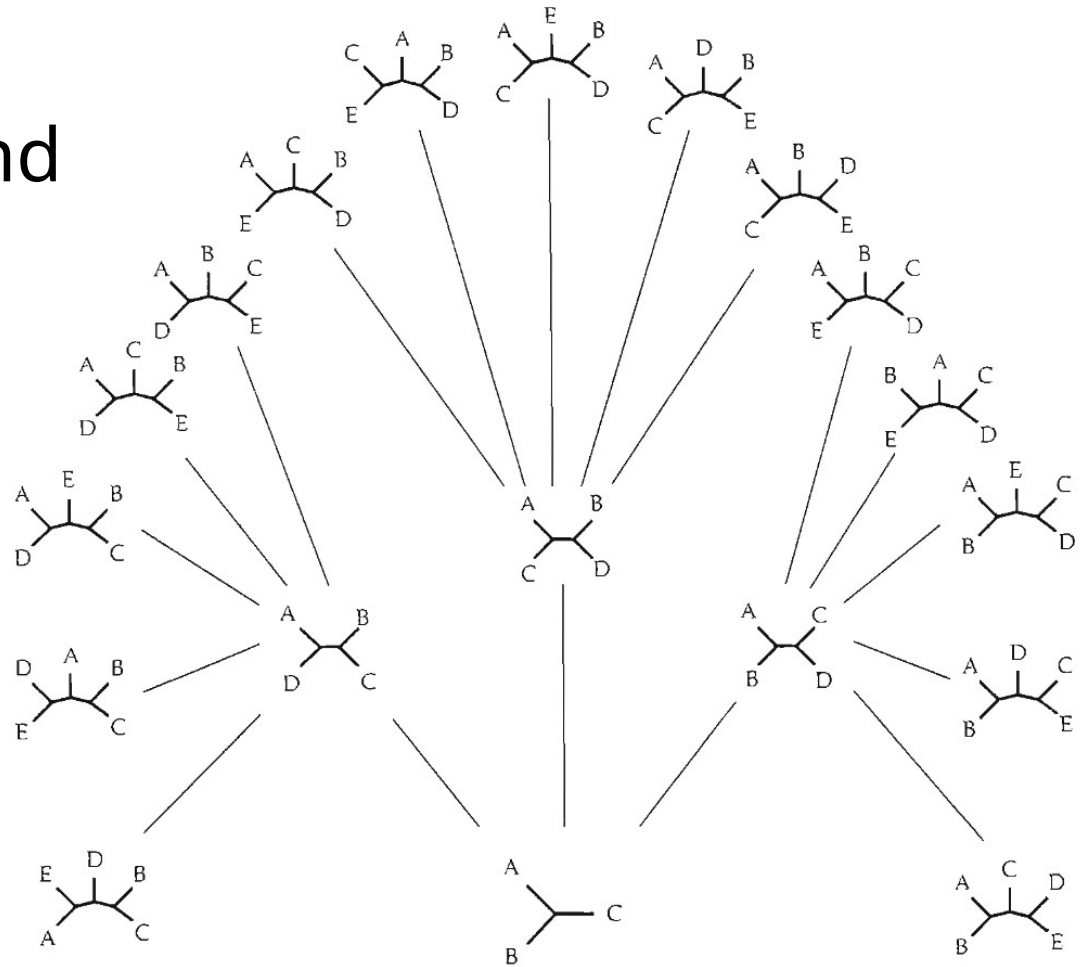


Figure 5.3: Search tree for most parsimonious tree in a five-species case.

Branch and bound

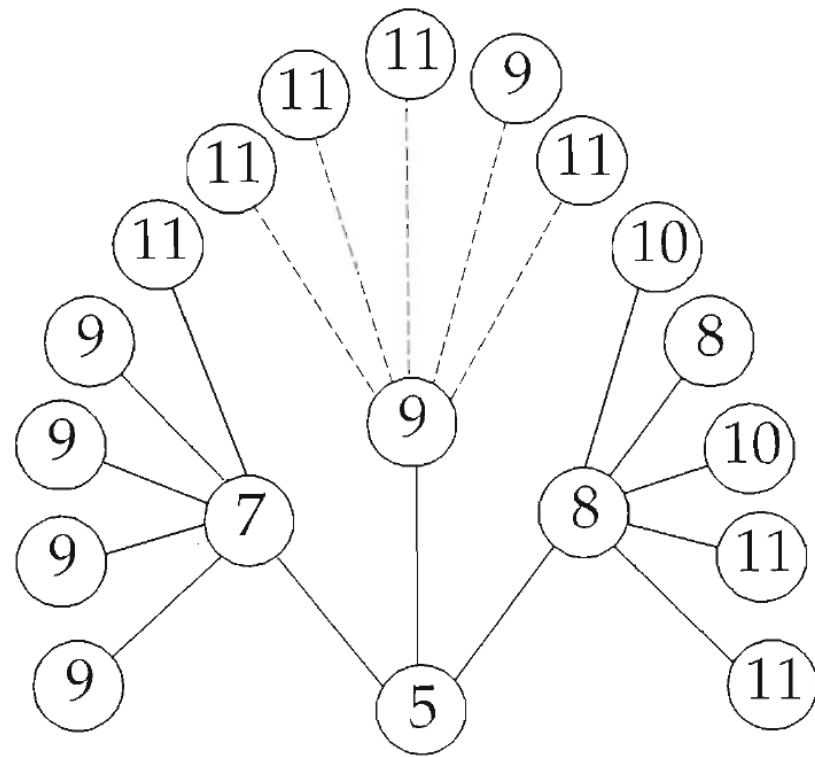


Figure 5.4: Search tree for most parsimonious tree for five species, using the data of Table 1.1. Trees are shown in Figure 5.3. Dashed lines are those not traversed by a branch and bound method. The species names in the data set correspond to labels A through E in Figure 5.3.